

# Proprioceptive Neuromuscular Facilitation

## PNF

## Introduction

- **Made up of 3 words :**
  1. **Proprioceptive**
  2. **Neuromuscular**
  3. **Facilitation**
- **Technique that combines different pattern of movements with techniques of neuromuscular facilitation to stimulate motor response and further improve neuromuscular control and function.**
- **PNF is an **integrated approach**.**
- **Primary goal is to achieve the patient's **highest level of function**. For this, therapist integrates principles of motor learning and motor control.**

## Principles & Procedures

- **Resistance**
- **Irradiation**
- **Manual Contact**
- **Position**
- **Verbal command**
- **Vision**
- **Traction**
- **Approximation**
- **Stretch**
- **Timing**
- **Patterns**

## Resistance

- **Goal**
  - Facilitate muscle's ability to contract
  - Increase motor control and learning
  - Gain awareness about motion and direction
  - Gain Strength
- The amount of resistance provided must be correct for patient's condition
- When a muscle contraction is resisted, muscle's response to cortical stimulation increases.
- Breathing should be maintained.
- Resistance should not cause any pain or unwanted fatigue.

## Irradiation

- **Spread of response to the stimuli.**
- **It can be contraction or relaxation**
- **Increasing amount of resistance will increase the amount and extent of muscular response.**
- **Eg:**
  - **Resist hip flexion to cause trunk flexor contraction**
  - **Resist supination to cause contraction of external rotators of shoulder**

## Manual Contact

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## Manual Contact

- The therapist's grip stimulates the patient's skin receptors.
- Therapist hand should be placed to apply pressure opposite to the direction of motion.
- Promotes **Tactile Kinesthetic Sense**
- To control movement and resist rotation, the therapist uses **lumbrical grip**.
- This gives therapist a good control over the patient's motion without any pain and discomfort
- Example :
  - **From standing to sitting** – Therapist can give patient a kinesthetic information for the movement by putting hands on iliac crest and applying pressure downward and backward.

## **Body Position**

- **Give therapist effective control of patient's motion**
- **Facilitate control of direction of resistance**
- **Enable therapist to give resistance without fatigue**
- **Therapist body should be aligned with desired motion.**
- **Therapist's shoulder and pelvis should face the direction of motion**
- **The resistance should come from the therapist's body while hands and arms will be comparatively relaxed.**



## Verbal Commands

- **Guide initiation of muscle contraction and movement.**
- **Evoke the motor function**
- **Timing of command is important to coordinate patient's reaction to therapist.**
- **Command is divided into 3 parts:**
  - 1. Preparation** – To make patient ready for action
  - 2. Action** – Tell the patient to start the action
  - 3. Correction** - Tell the patient to correct the movement/position

## Vision

- **Promote more powerful muscle contraction.**
- **Helps the patient to correct and control the position**
- **Influence both head and body movement**
- **When patient looks at his/her movement, a stronger feedback is achieved**

## Traction

- **Facilitate motion**
- **Resist some part of motion**
- **Traction is maintained throughout the movement and combined with appropriate resistance.**
- **Example : Traction at beginning of shoulder flexion resist scapula elevation**

## Approximation

- **Promote stabilization**
- **Facilitate weight bearing and contraction of antigravity muscles**
- **Traction facilitates motion and Approximation facilitates stability, the therapist should use the one which is more effective.**

## Stretch

- **Facilitate muscle contraction**
- **Activates stretch receptor**
- **The stretch stimulus also facilitates synergistic and other associated muscles**
- **Example – Elongation of anterior tibial muscle facilitates it and also facilitates Hip Flexor, Adductor and External rotator muscle group**

## Timing

- **Movement should be done from distal to proximal**
- **Normal movement requires a smooth sequence of activity and coordinated movement requires precise timing of that sequence.**